

NATIONAL RESEARCH UNIVERSITY
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Research Project Report on the Topic:
A/B Testing of a Recommender System as a Black-Box Experiment

Submitted by the Student:

group #БПА/Д244, 2nd year of study Sapunov Artemiy Maximovich

Approved by the Project Supervisor:

Ramazyan Tigran Armenovich
Research Fellow
Faculty of Computer Science, HSE University

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Abstract

This project presents a complete black-box A/B testing pipeline applied to a UI button experiment. The joint system of UI element and user is treated as a black box: the input is the assignment of a user to control variant A or treatment variant B, and the outputs are product metrics (click-through rate, CTR) computed from logged user events. The work delivers a formalised experimental methodology (unit of randomisation, metric hierarchy, statistical analysis plan), a calibrated simulator grounded in real data, and a black-box parameter optimisation component that searches over button configuration space to maximise CTR. The project addresses common practical problems including sample ratio mismatch (SRM), the choice between per-user and per-impression CTR definitions, and the cost of black-box optimisation.

Аннотация

Проект представляет собой полный пайплайн А/В-тестирования в постановке «чёрного ящика», применённый к эксперименту с UI-кнопкой. Система рассматривается как чёрный ящик: входом является назначение пользователю варианта А (control) или В (treatment), а выходом --- продуктовые метрики (CTR), рассчитанные по логам событий. Результатом работы являются формализованная методология эксперимента, откалиброванный симулятор, основанный на реальных данных, и компонент оптимизации параметров кнопки. Рассматриваются типовые проблемы: SRM, выбор метрики CTR и стоимость оптимизации чёрного ящика.

Keywords

A/B testing; online controlled experiments; black-box simulation; UI button experiment; CTR; per-impression CTR; SRM; black-box optimisation; SymPy; calibration

1 Introduction

1.1 Subject Area and Motivation

Modern digital products improve user experience through stepwise changes to UI elements, ranking logic, and recommendation features. Online randomised experiments (A/B tests) are the most reliable decision instrument for evaluating such changes, because they support causal interpretation of the observed effect. This project uses a UI button change as the experimental object, treating the button–user system as a black box to study the input–output relationship in isolation from the internal mechanism.

1.2 Problem Statement

We consider a webpage with a button whose visual parameters (colour, size, label text) can be varied continuously in $[0, 1]$. A new button configuration (version B) is proposed. The goal is to determine whether version B improves the primary product metric (CTR) compared to the current version A, and, given a calibrated simulator, to identify the configuration that maximises CTR.

Statement. Run an A/B test where users are randomly assigned to A or B. Using event logs, compute agreed metrics and decide whether to deploy B. Additionally, query the calibrated black box to find the parameter combination that maximises expected CTR.

1.3 Outcomes

The project delivered:

- a formalised description of the experiment, including the unit of randomisation, metric hierarchy, and statistical analysis plan;
- a Python package `ab_blackbox` with symbolic response models (SymPy), a black-box simulator, a statistical analysis pipeline, and dataset calibration utilities;
- a real-data analysis on a publicly available 120 000-user dataset, yielding a statistically significant +10.0% CTR lift and a **SHIP** decision;
- a black-box optimisation component comparing grid search, random search, and greedy sweep, with results averaged over 20 seeds.

2 Literature Review

The project draws on two bodies of literature: causal inference foundations and practical online experimentation.

Causal foundations. Rubin (1999) establishes the potential outcomes framework and shows that random assignment is sufficient for causal interpretation of observed differences [6]. This paper provides the theoretical basis for interpreting the A/B test as a causal experiment rather than a mere correlation study. Our project adopts Rubin’s framework explicitly in the Assumptions section: SUTVA, ignorability, and positivity are stated as preconditions, and all three are satisfied by design under user-level randomisation with a 50/50 split.

Practical online A/B testing. Kohavi and Longbotham (2016) describe the end-to-end lifecycle of online controlled experiments in production systems, including variant definition, randomisation unit selection, and guardrail metrics [5]. Deng et al. (2017) extend this with statistical warnings: naive analysis that ignores the randomisation unit underestimates variance and overstates significance [1]. Our project follows both papers by fixing the unit at the user level, using binary per-user CTR as the primary metric, and running a supplementary per-impression CTR analysis to confirm that the result is not an artefact of the metric definition.

Pitfalls and diagnostics. Johari et al. (2017, 2022) show that continuous monitoring invalidates classical hypothesis tests and propose always-valid inference alternatives [4, 3]. We address this by using a fixed-horizon stopping rule: no interim analysis is performed, which keeps the type-I error at the nominal $\alpha = 0.05$. Fabijan et al. (2019) demonstrate that sample ratio mismatch (SRM) is a common and underdiagnosed data quality problem that can invalidate all downstream analysis [2]. Following their recommendation, our pipeline runs an SRM chi-square check before any metric analysis.

Implementation Overview

The practical component is a Python package `ab_blackbox` with five modules and three scripts. Table 1 lists all files and their roles.

Table 1 — Project files and their roles

File	Layer	Role
<code>ab_blackbox/model.py</code>	Package	Symbolic response models (SymPy)
<code>ab_blackbox/simulator.py</code>	Package	Black-box oracle
<code>ab_blackbox/experiment.py</code>	Package	A/B experiment runner
<code>ab_blackbox/analysis.py</code>	Package	Statistical analysis pipeline
<code>ab_blackbox/datasets.py</code>	Package	Data loading and calibration
<code>ab_blackbox/__init__.py</code>	Package	Public API
<code>demo.py</code>	Script	Synthetic end-to-end demonstration
<code>main.py</code>	Script	Real-data pipeline
<code>optimize.py</code>	Script	Black-box parameter optimisation
<code>tests/test_ab_blackbox.py</code>	Tests	14 unit tests

3 Experimental Design

The experiment follows a standard two-variant randomised design. Users are assigned independently and uniformly at random to control (A) or treatment (B) at the user level, ensuring consistent exposure throughout the experiment.

The *experimental object* is a button displayed to users on a webpage. The treatment modifies three button parameters—colour, size, and label text—each encoded as a continuous value in $[0, 1]$, where 0 is the current default and 1 is the fully activated alternative. The baseline experiment sets `color` = 1 while holding `size` = `text` = 0, matching the real-world dataset structure where only one change separates control and test.

The joint system of button and user is treated as a *black box*: the only observable quantities are the variant assignment and the output event log. A single call to the black box simulates the full experiment for a given parameter configuration and returns aggregate metrics. Call count is tracked to quantify the cost of optimisation.

The dataset used for calibration contains $n = 60,000$ users per group, substantially exceeding the analytically derived minimum of $n = 9,458$ required to detect the observed effect at 80% power and $\alpha = 0.05$ (Section 6).

4 Metric System

Three tiers of metrics are defined prior to data collection.

Primary metric. Per-user click-through rate (CTR), defined as the fraction of users who clicked the button at least once:

$$\widehat{\text{CTR}}_{\text{user}} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{clicks}_i \geq 1].$$

This is a conservative definition that counts each user at most once, regardless of how many clicks they made.

Supplementary metric. Per-impression CTR, defined as the ratio of total clicks to total page views across all users:

$$\widehat{\text{CTR}}_{\text{impression}} = \frac{\sum_{i=1}^n \text{clicks}_i}{\sum_{i=1}^n \text{views}_i}.$$

This metric uses the full click count and normalises by opportunity (views). It is reported alongside the primary metric to confirm that the result is not an artefact of the binary per-user definition. The two metrics are consistent: per-user lift +10.0%, per-impression lift +8.3%.

Guardrail metric. The sample ratio $\hat{r} = n_A / (n_A + n_B)$ is monitored before any metric analysis. A significant deviation from 0.5 (chi-square test at $\alpha = 0.01$) invalidates the experiment.

5 Data Quality and Logging

The dataset `ab_test_results_aggregated_views_clicks_2.csv` contains 120,000 rows with columns `user_id`, `group`, `views`, and `clicks`. Groups are `control` and `test`. The dataset is perfectly balanced ($n_A = n_B = 60,000$), has no missing values, and no user has more clicks than views. No SRM is present ($p = 1.000$).

The `views` column records per-user page-view counts (range 1–205, mean ≈ 5). It is used to compute the per-impression CTR. The `clicks` column records raw click counts per user (range 0–9); binarised to $\mathbf{1}[\text{clicks} \geq 1]$ for the per-user CTR.

6 Statistical Analysis Plan

Primary test. The treatment effect on per-user CTR is estimated using a two-proportion z -test. Let p_A and p_B denote the true click probabilities. The null hypothesis is $H_0 : p_B = p_A$ against the two-sided alternative $H_1 : p_B \neq p_A$. The test statistic is

$$z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}, \quad \hat{p} = \frac{x_A + x_B}{n_A + n_B},$$

where x_A, x_B are click counts. The 95% CI on $\Delta = p_B - p_A$ uses the unpooled standard error:

$$\Delta \pm z_{0.025} \sqrt{\frac{\hat{p}_A(1 - \hat{p}_A)}{n_A} + \frac{\hat{p}_B(1 - \hat{p}_B)}{n_B}}.$$

Minimum sample size.

$$n = \frac{(z_{\alpha/2} + z_{\beta})^2 (p_A(1 - p_A) + p_B(1 - p_B))}{\delta^2},$$

where $\delta = |p_B - p_A|$. Derived symbolically in `model.py` using SymPy and evaluated at the calibrated parameter values.

Bootstrap robustness check. A non-parametric bootstrap CI (2000 resamples, vectorised) is computed alongside the z -test. Agreement confirms the z -test assumptions are satisfied.

Stopping rule. The experiment runs for a fixed pre-determined duration. No interim analysis is performed, which avoids the inflated type-I error from continuous monitoring [4].

7 Risks and Limitations

Novelty effect. Users may click the new button simply because it is unfamiliar. The dataset contains no timestamps, so novelty decay cannot be measured directly. Monitoring post-

deployment CTR would confirm whether the lift persists.

Assumed parameter weights. The dataset contains only two conditions (color=0 vs. color=1). The weights for **size** and **text** are not observed and are set equal to the colour effect ($w = 0.0147$) as a *baseline assumption with no empirical basis*. The optimisation results are therefore illustrative only; follow-up experiments varying size and text are needed to replace the assumption with measured values.

Per-user binary CTR. Using $\mathbf{1}[\text{clicks} \geq 1]$ treats a user with one click identically to a user with nine clicks. The per-impression metric addresses this, and both tell a consistent story.

Simulation noise. The simulator adds Gaussian noise to p_{click} once per black-box call (noise_std= 0.003 for optimisation). This models between-experiment variation in population CTR, not within-experiment confounding. For calibration validation, noise_std= 0 is used so that the simulated Δ matches the calibration target exactly.

8 Decision Rules

The decision framework has four outcomes:

1. **INVALID:** SRM detected ($p < 0.01$). Halt; investigate.
2. **SHIP:** Primary metric significantly positive. Deploy B.
3. **REJECT:** Primary metric significantly negative. Do not deploy.
4. **INCONCLUSIVE:** Not significant. Increase sample size or duration.

9 Assumptions

The analysis rests on four assumptions from the potential outcomes framework [6]:

1. **SUTVA:** Each user’s outcome depends only on their own assignment (no interference between users).
2. **Ignorability:** Assignment is independent of potential outcomes. Satisfied by design under random assignment.
3. **Positivity:** Each user has a non-zero probability of assignment to either variant. Satisfied by the 50/50 split.
4. **Stable response surface:** Click probability is modelled as a deterministic function of button parameters plus i.i.d. noise. Parameter interactions are not modelled.

Experimental Results

Table 2 reports key metrics for the direct test on real data and the calibrated simulation (noise_std= 0).

Table 2 — A/B test results: real data and calibrated simulation

Source	CTR A	CTR B	Δ	p -value
Real data (direct z -test)	0.1470	0.1617	+0.0147	< 0.0001
Calibrated simulation (noise_std= 0)	0.1462	0.1623	+0.0161	< 0.0001

For the direct test: $z = 7.055$, 95% CI on Δ : [+0.0106, +0.0188], relative lift +10.0%. The per-impression CTR also increased significantly (+8.3%, $z = 5.52$, $p < 0.0001$), confirming that the result is not an artefact of the per-user binary definition.

With noise_std= 0, the calibrated simulation’s Δ (+0.0161) is close to the real Δ (+0.0147); the small discrepancy is due to Bernoulli sampling variance with seed 42 at $n = 60,000$. The bootstrap CI [+0.0121, +0.0204] excludes zero and agrees directionally with the z -test. No SRM was detected ($p = 1.000$). The decision rule yields **SHIP**.

Required sample size. The minimum per-group n to detect $\delta = 0.0147$ at 80% power and $\alpha = 0.05$ is $n = 9,458$, derived analytically from the SymPy model. The actual dataset ($n = 60,000$) is approximately $6\times$ larger than required.

Black-box optimisation. Table 3 shows results averaged over 20 independent seeds. All three methods find the same optimal configuration (color = 1, size = 1, text = 1); the difference in mean CTR reflects the efficiency of each search strategy, not a difference in the configuration found. The theoretical CTR at (1, 1, 1) from the SymPy model is 0.1912.

Table 3 — Optimisation results averaged over 20 seeds. All methods find config (1, 1, 1); CTR differences reflect sampling noise, not different optima. The theoretical CTR at (1, 1, 1) is 0.1912 (SymPy).

Method	Calls	Mean CTR	Std	Lift vs. real test
Real experiment (A→B)	—	0.1617	—	(reference)
Grid search	125	0.1973	0.0038	+22.0%
Random search (30 calls)	30	0.1917	0.0037	+18.5%
Greedy 1-D sweep	15	0.1890	0.0061	+16.9%
SymPy theoretical	0	0.1912	—	+18.2%

Discussion: Interpretation and Effects

Treatment effect. The test variant produces a statistically significant increase in per-user CTR of +0.0147 (relative lift +10.0%) and in per-impression CTR of +0.0025 (+8.3%). The effect is robust: confirmed by the parametric z -test, the non-parametric bootstrap, and two complementary metric definitions.

Black-box perspective. Treating the button–user system as a black box is methodologically appropriate: no assumptions about the internal mechanism are required. The causal interpretation relies only on random assignment [6], which is satisfied by design.

Optimisation vs. single test. The standard A/B test compares two fixed configurations. The optimisation component extends this to a search over the three-dimensional parameter space. Averaging over 20 seeds shows that random search (30 calls) recovers a mean CTR of 0.1917, compared to grid search’s 0.1973 (125 calls), illustrating the efficiency gain from sub-exhaustive methods. The greedy sweep uses only 15 calls but has higher variance ($\sigma = 0.0061$), since it can miss parameter interactions.

The SymPy theoretical value (0.1912) serves as the ground truth: it is the true model maximum at (1, 1, 1), computed with zero black-box calls and zero noise. Any method’s mean CTR that exceeds 0.1912 reflects noise, not a superior algorithm.

Limitations. The weights for `size` and `text` are not observed from data and are set to a baseline assumption equal to the colour weight. Conclusions about the optimal multi-parameter configuration depend on this assumption. A follow-up experiment would allow the model to be re-calibrated with observed weights, making the optimisation results empirically grounded.

Conclusion

This project delivered a complete black-box A/B testing pipeline, from symbolic response modelling (SymPy) through simulation, statistical analysis, and parameter optimisation. The treatment variant produces a statistically significant +10.0% lift in per-user CTR (+8.3% per-impression), the decision rule yields **SHIP**, and the optimisation component—averaged over 20 seeds—identifies configurations with up to +22.0% additional lift using 125 calls (grid search) or +18.5% with 30 calls (random search).

References

- [1] Alex Deng, Jiannan Lu, and Jennifer Litz. “Trustworthy Analysis of Online A/B Tests: Pitfalls, Challenges and Solutions.” In: *Proceedings of the Tenth ACM International Conference on Web Search and Data Mining*. 2017.
- [2] Aleksander Fabijan, Pavel Dmitriev, Jay Gupchup, Jason Omhover, and Lukas Vermeer. “Diagnosing Sample Ratio Mismatch in Online Controlled Experiments.” In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2019.
- [3] Ramesh Johari, Peter Koomen, and Leonid Pekelis. “Always Valid Inference: Continuous Monitoring of A/B Tests.” In: *Operations Research* (2022).
- [4] Ramesh Johari, Peter Koomen, Leonid Pekelis, and David Walsh. “Peeking at A/B Tests: Why It Matters, and What to Do About It.” In: *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2017.
- [5] Ron Kohavi and Roger Longbotham. “Online Controlled Experiments and A/B Testing.” In: *Encyclopedia of Machine Learning and Data Mining*. Springer, 2016.
- [6] Donald B. Rubin. “Teaching Causal Inference in Experiments and Observational Studies.” In: *Proceedings of the American Statistical Association, Section on Statistics Education* (1999).